NHL Data Mining: Optimal Strategy on a Man Advantage or Disadvantage

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Abstract: This study investigates two critical aspects of professional ice hockey in the National Hockey League (NHL): the effectiveness of power play and penalty kill units, and the optimal timing for pulling a goalie during last-minute urgency. A novel player effectiveness rating system is developed, based on situational statistics weighted according to players' roles and positions. These ratings are discretized into five levels using normalization and equal-width binning for enhanced interpretability. To identify attributes of successful special teams, unsupervised clustering and supervised machine learning models are applied, revealing player weight as a key predictor of power play and penalty kill success. Additionally, an interpretable decision tree model is constructed to evaluate the impact of goalie pulling timing on game outcomes. The findings suggest that the optimal time to pull the goalie is between 1:42 and 2:35 in the third period. These insights provide practical guidance for coaching strategies, emphasizing the importance of physical attributes and timely decision-making in critical game situations.

I. Introduction

The National Hockey League (**NHL**) is the premier hockey league in the world, featuring the highest level of talent from across the globe. The NHL consists of 32 teams, all of which host their games in North America, primarily Canada and the United States. While the majority of NHL fans reside in these countries, there is a significant global fan base that enjoys the sport of hockey. Most NHL games are played in the cities where the teams are based, but occasionally, teams travel to other countries for the NHL Global Series. For example, in the upcoming 2025-2026 NHL season, the Pittsburgh Penguins will face the Nashville Predators in Sweden as part of the NHL Global Series [1].

This study examines strategic plays in hockey, focusing on situations where a team has an advantage or disadvantage. In the NHL, when a team draws a penalty, it has the opportunity to play to an advantage, known as a *power play*. The duration of penalties can vary, with minor penalties lasting two or four minutes and major penalties lasting up to five minutes. Being on the power play has been shown to be an effective way for teams to outperform their opponents. Teams that accumulate more penalty minutes and spend more time at a disadvantage generally perform significantly worse than those who enjoy the advantages of the power play.

In the NHL, a common last-minute strategy involves a team replacing its goalie with an extra skater to gain an offensive advantage. This tactic usually occurs near the end of the game when a team is trailing. By pulling the goalie, the team increases its chances of scoring to tie the game and possibly force it into overtime. However, if a team pulls their goalie too early, they risk allowing the opposing team to score into the empty net since there's still plenty of time left. Conversely, if they wait too long to make the change, they may have insufficient time to score and improve their chances of winning.

This work investigates the advantages and disadvantages of teams in order to gain insights into what contributes to effective power plays and penalty kills. The analysis focuses on teams during critical moments, particularly the last-minute desperation play, by examining the players on the ice and the timing of when goalies are pulled. Understanding the factors that lead to improved performance can help coaching staff make better decisions regarding line formations, roster selections, and strategies, thereby positively influencing their teams' overall performance. The following research questions are addressed to achieve these insights:

1) What makes a **successful power play or penalty kill**?

2) What is the **optimal goalie pull time** to force overtime?

II. Related Works

Macdonald [2] utilized a range of NHL statistics, such as goals, missed shots, blocked shots, turnovers, and face-off wins, to develop models that predict a player's expected goals. To address the issue of collinearity in the data, ridge regression was employed, which helps to reduce error bounds in the estimates. This approach enhanced the predictive power of the models. Three seasons worth of data were used to train the models, while one season was reserved for validation. An even-odd split of games was implemented to account for mid-season trades and injuries that could affect team performance, where statistics from odd games were used to predict outcomes in even games.

Chin et al. [3] developed models using Decision Trees and Artificial Neural Networks to predict the outcomes of NHL games. The data for their study were obtained from a research team at the University of Ottawa, which covered a total of 517 games, resulting in 1034 records since two entries were included for each team. Each record compiled various team statistics leading up to the game, including statistics such as goals for, goals against, power play and penalty kill win percentages, and win streaks. The primary prediction target was whether the team emerged victorious in the game.

Bunker and Thabtah [4] developed a sports prediction framework utilizing Machine Learning techniques. Their work outlines the methodologies for comprehending the sport, managing the data, and constructing models within the field of sports analytics. They analyzed the methods employed and their application in sports analytics for each step of the framework. The modeling techniques explored include Artificial Neural Networks, noted for their impressive predictive capabilities. They define supervised and unsupervised learning models, demonstrating how classification, regression, and clustering tasks can be effectively applied.

Our work tackles specific times of the game, focusing on effective power plays and penalty kills to find optimal strategies for handling these situations.

III. Data

Our research collected most of our data from a Kaggle dataset [5] that encapsulated NHL game statistics from 2000 to 2019. Additionally, we obtained valuable information about player positions for each season from MoneyPuck [6]. To refine our analysis, we decided to focus on data from 2010 onward, as crucial statistics relevant to our study—such as player shifts during games—were not documented before this period. We narrowed our focus to 5-on-4 situations for penalty kills and power plays, and 6-on-5 situations when the team was trailing by one goal for goalie pulls. The NHL is highly situational, with various combinations of players on the ice during advantageous and disadvantageous scenarios. The most common occurrences, such as 5-on-4 power plays, penalty kills, and 6-on-5 goalie pulls, provided us with a substantial dataset for our analysis.

The relational database from Kaggle provided us with 13 tables. Not all the tables in the database were utilized, as some contained information irrelevant to our research. Tables that are not utilized include game scratches, game officials, and team info. Game scratches included all players absent during a game, which we did not need to consider as we were considering the players who played in the games. The game officials dataset provided all referees who officiated each game, and this data was not valuable for our research. Finally, the team info dataset provided basic naming conventions for each team in the NHL, such as their city/state name and team name, as well as an abbreviation for each team.

As a result of narrowing down the data and putting aside irrelevant information, we were left with 10 total tables to collect and prepare the data for use. The following table displays each dataset with the number of dimensions:

Dataset	# of Rows	# of Cols.
Game	26,305	15
Game Goals	148,992	4
Game Penalties	247,828	3
Game Plays	5,050,529	18
Game Plays Players	7,586,604	4
Game Shifts	11,882,735	5
Game Skater Stats	945,830	22
Game Goalie Stats	56,656	19
Game Team Stats	52,610	17
Player Info	3925	12

To conduct our research effectively, we needed to extract significant information from the data available. To achieve this, we employed feature creation for essential statistics such as goalie pull times and

line changes, which were not readily included in our datasets. This approach helped us mitigate the curse of dimensionality and concentrate on the most relevant features for our study. We also applied binning techniques to several columns to facilitate analysis, particularly during clustering and classification, yielding more meaningful results. The features analyzed included players' height, weight, position, and more. Notably, we developed key features such as Players on Ice and new player evaluation metrics.

A. Deriving Players on Ice

To identify the players on the ice during power plays and penalty kills, we utilized the shifts table from the Kaggle dataset, which offers timestamps for each player's shifts within a game. By cross-referencing these shift times with the timing of penalties, we could ascertain the precise lineups during 5-on-4 situations. A 5-on-4 situation occurs when a team is penalized during a play on even strength (5-on-5), and the time assessed for the penalty is played at a disadvantage. The team that draws the penalty plays with their usual five players, while the team that was penalized plays with four players. The penalized player gets placed in the penalty box, where they sit and wait for the penalty's allocated time. Upon completing the 5-on-4 situation, the player in the box is allowed out, where the situation changes to even strength.

1) Power Play and Penalty Kill

We focused on two-minute minors to analyze 5-on-4 power plays, using penalty start and end times to define our analysis windows. We identified players with overlapping shifts during these windows to reconstruct the on-ice configurations for both power play and penalty kill units. When a goal occurs during a penalty, we noted the players on the ice, which allowed us to examine the combinations involved in successful and unsuccessful power plays.

Since the data didn't explicitly label power play or penalty kill units, we needed to use additional logic to assign roles. We determined which team was on the power play by counting the number of skaters on the ice during the 5-on-4 window, where the team with five players was assigned to the power play and the team with four to the penalty kill. We repeated this process across all games, accounting for irregularities like overlapping and coincidental penalties while maintaining a clean dataset of standard 5-on-4 situations. This enabled us to analyze special teams performance at the player-combination level.

2) Goalie Pulls

Due to the absence of goalie pull times in the available data, we performed data processing to derive these times for our research. The game shifts table in the database includes all player shifts during games, along with their corresponding game IDs and player IDs. The key columns were the game period, shift start, and shift end. By identifying when a goalie commenced and concluded their shift, we were able to determine the times when the goalie left the ice. Our ultimate objective was to obtain intervals during which the goalie was off the ice, enabling us to retrieve the players who were on ice at that time. We defined the start of each interval as the goalie's pull time and utilized the entire interval to identify the players active during that play.

Determining the intervals during which a goalie is off the ice requires careful consideration of the game dynamics. A goalie leaving the ice does not always account for those last-minute desperation plays. There are several reasons a goalie might exit, such as being replaced due to subpar performance or injury, the conclusion of a shift at the end of a period, or a strategic decision to pull the goalie during a delayed penalty. This delayed penalty occurs when a player on the opposing team commits a foul. The stoppage of play happens when the penalized team touches the puck, allowing the opposing team the opportunity to play with an empty net. In such situations, teams often choose to pull their goalie, substituting them with an extra attacker to capitalize on the advantage during the delayed penalty.

Taking all factors into account, additional data cleaning was performed, narrowing our focus to only last-minute desperation plays. The dataset was initially filtered to include only goalies, as it originally contained information on all players. This was achieved by using player IDs alongside the game goalie stats table, which features only goalie data. Consequently, a team ID column was created to differentiate between goalie teams. Since the plays occurred solely in the third period, we limited the dataset to shifts within this time frame. As noted earlier, goalies with a shift end time of 3600 seconds (the end of the third period) were excluded, as this indicated the end of the period rather than a shift change. To simplify further cleaning, the data was organized by game ID, shift start, and team ID in that order. These data cleanup steps formed the foundation for our examination of delayed penalty pulls and goalie replacements.

The replacement of goalies due to poor performance or injury does not qualify as a typical goalie

pull for last-minute desperation plays. This distinction became evident from the data, which indicated that when a goalie left the ice, the subsequent shift recorded a different player from the same team. Therefore, shifts that concluded with a different player were excluded. Additionally, shifts that ended close to a penalty call on the ice were also eliminated to avoid confusion with goalie pulls resulting from the delayed penalty scenario. After accounting for these factors, the final intervals of goalies' time off the ice were calculated.

With access to accurate goalie time-off ice intervals, we could identify the players present on the ice during these moments. This retrieval involved simply collecting the players active during the time-off ice interval from the game shifts table. Consequently, for each coach's initial plan to pull the goalie, six players were documented for the advantage team and five for the disadvantaged team. If the advantage team scored, thus equalizing the game and necessitating overtime, the players on the ice at the time of the goal were recorded. The resulting dataset included the goalie pull time, the outcome of the game (Overtime or Regulation Time), the players involved, and additional player statistics such as position, height, weight, handedness, and nationality.

B. Evaluating Player Effectiveness

Dom Luszczyszyn [7] is a prominent NHL writer for The Athletic, with a focus on hockey analytics. His previous work [8] introduced an innovative method for assessing player performance through the development of a new metric. He created a framework to evaluate individual player performance on a game-by-game basis using a Game Score metric. This metric has enhanced our understanding of individual player contributions, particularly in instances where performance evaluations might be ambiguous or subject to debate.

Statistics used to evaluate a player's performance include goals (G), primary assists (A1), secondary assists (A2), shots on goal (SOG), blocks (BLK), penalties drawn (PD), penalties taken (PT), faceoff wins (FOW), faceoff losses (FOL), Corsi for (CF), Corsi against (CA), goals for (GF), and goals against (GA). A goal occurs when a skater puts the puck into the net and gains a point for their team score. The primary assist is the second last player to touch the puck due to the goal, and the secondary assist is the third last player to touch the puck. A shot on goal is a shot taken by a player that results in a save by the goalie or a goal. Blocking a shot occurs when a skater successfully blocks a shot in their opponent's attempt at a shot on goal. Taking a penalty is the player that gets penalized, while drawing a penalty is the recipient player of this play. At the start of resuming the game following a stoppage, players line up to take a 1-on-1 faceoff where the referee will drop the puck, giving each team an equal opportunity to gain possession. Winning faceoffs results in a player gaining possession for their team while losing faceoffs results in losing possession. Corsi is defined by the number of shot attempts taken, which includes shots on goal, shots that were missed, and shots that were blocked. Corsi for is the statistic for a player when they are taking the shots, and Corsi against is when the player gets shot on. Similarly, goals for are the statistics when a player scores and goals against are when the player gets scored on. Below is the formula for Player Game Score:

$$\begin{aligned}\text{Game Score} = \\ (0.75\text{*G}) + (0.7\text{*A1}) + (0.55\text{*A2}) + \\ (0.075\text{*SOG}) + (0.05\text{*BLK}) + (0.15\text{*PD}) - \\ (0.15\text{*PT}) + (0.01\text{*FOW}) - (0.01\text{*FOL}) + \\ (0.05\text{*CF}) - (0.05\text{*CA}) + (0.15\text{*GF}) - (0.15\text{*GA})\end{aligned}$$

Evaluating a player's effectiveness became a crucial step in determining how effectively a player can perform on the man advantage or disadvantage. By knowing how effective a player can be, we know how much they contribute to the overall outcome with their time on ice. No single statistic in the NHL can determine how well a player performs on the man advantage or disadvantage. A common statistic assigned to a player is their Plus-Minus **+/-**. This performance metric is an accumulated summation of the goals scored while the player is on the ice. Simply, if a player is on the ice and their team scores, their +/- increases by one, and if a player is on the ice and their team gets scored on, their +/- decreases by one. The +/- statistic can be used to evaluate a player's performance, but it can be misleading, and by no means is it perfect. This performance metric is not necessarily a measure of their individual performance but of their performance on the ice as a team. For example, if a player on the ice had no impact on the play and the goal result, this player would still be increasing their metric. Considering how the man advantage or disadvantage is played, we created a new feature to represent how effectively a player performs using a combination of statistics at an individual and team level.

The first step in creating this new feature was to single out players' positions and define the roles they have on the ice. Different player positions will have

different statistics, specifically the role of taking the faceoff for centermen. Being on the man advantage should also consider different statistics compared to the disadvantage. On the man advantage, the effectiveness should result from more offensive statistics, while the disadvantage should result from more defensive statistics.

All effectiveness formulas are similar, with some statistics considered only when applied to the player's position. Statistics in the formulas include faceoff win percentage (FOW%), goals (G), assists (A), plus-minus (+/-), shots on goal (SOG), giveaways (GV), blocks (BLK), takeaways (TKA), and hits (H). Most of these statistics, such as Plus-Minus, are explained previously in the paper, and similar statistics are used in the game score metric. Instead of separating primary and secondary assists, our assists represent both primary and secondary. As opposed to using the faceoff win-loss differential used in the game score metric, we used the faceoff win percentage, which is calculated by dividing faceoff wins by the total faceoffs taken. A giveaway occurs when a player loses possession of the puck when their opponent takes it away, and a takeaway occurs when a player takes the puck away from their opponent. Hockey is a contact sport that allows players to hit their opponents using their upper bodies, which results in a hit for a player. The following is a list of the formulas and a few details:

The most important difference is the centerman's role in successfully winning faceoffs. Winning faceoffs solidifies the team's control of the puck at the start of the play, which is crucial to a man advantage. The formula consists mainly of offensive statistics as well as subtraction in giveaways.

$$\text{Center Advantage Effectiveness} = \frac{w(\text{FOW\%}) + w(\text{G}) + w(\text{A}) + w(\text{+/-}) + w(\text{SOG}) - w(\text{GV})}{\text{Total Time on Advantage}}$$

The winger gets the faceoff win percentage removed. Although there are some cases where a winger could take faceoffs due to a false start in the team's first attempt at the faceoff, generally, the centerman takes this role, and wingers stay to the left and right sides of the draw.

$$\text{Winger Advantage Effectiveness} = \frac{w(\text{G}) + w(\text{A}) + w(\text{+/-}) + w(\text{SOG}) - w(\text{GV})}{\text{Total Time on Advantage}}$$

The defensemen also get the faceoff win percentage removed as they rarely take this role. Defensemen also have their +/- removed because, for some players, the statistic was overpowering the numerator, resulting in the calculation becoming negative. It was overpowering because defensemen tend to score and assist far less than any other position.

$$\text{Defensemen Advantage Effectiveness} = \frac{w(\text{G}) + w(\text{A}) + w(\text{SOG}) - w(\text{GV})}{\text{Total Time on Advantage}}$$

The following formulas represent the evaluation of centermen, wingers, and defensemen on the disadvantage. These formulas are identical to the advantage formulas in terms of offensive statistics. In addition to offensive statistics, giveaways are replaced with defensive statistics like hits, blocks, and takeaways.

$$\text{Center Disadvantage Effectiveness} = \frac{w(\text{FOW\%}) + w(\text{G}) + w(\text{A}) + w(\text{+/-}) + w(\text{BLK}) + w(\text{TKA}) + w(\text{H})}{\text{Total Time on Disadvantage}}$$

$$\text{Winger Disadvantage Effectiveness} = \frac{w(\text{G}) + w(\text{A}) + w(\text{+/-}) + w(\text{BLK}) + w(\text{TKA}) + w(\text{H})}{\text{Total Time on Disadvantage}}$$

$$\text{Defensemen Disadvantage Effectiveness} = \frac{w(\text{G}) + w(\text{A}) + w(\text{BLK}) + w(\text{TKA}) + w(\text{H})}{\text{Total Time on Disadvantage}}$$

Statistics a player acquires can change during the season, and their total stats towards the end do not accurately represent the player at the beginning of the season. To represent a player's effectiveness by only the statistics they accumulate until the game they are currently playing, the effectiveness is an accumulation of statistics from the start of the available data we have until the current game they are playing. The statistics are specific to those acquired on the man's advantage or disadvantage. Although it is ideal to use statistics acquired from the 6-on-5 for goalie pulls, this last-minute desperation play is a rare occurrence, and using only these statistics would limit the knowledge gained. All statistics used are from when the player acquires them on the man advantage or disadvantage, with the only exception being goalie pulls advantage or disadvantage statistics acquired in any situation. Statistics on the numerator are assigned a weight as different statistics have different importance regarding their performance. For example, a goal should be assigned

a higher weight than an assist because a goal is more important than the pass that created the goal. As a result of the formula, as players acquire more important statistics in the numerator and spend less time on the ice, they will be evaluated as better than their teammates and opponents.

C. Player Ratings and Discretization

After evaluating the players, we needed to represent this continuous variable as a more understandable variable. Developing a rating system contributed to representing the players more clearly. The first step in developing this rating system was to simplify the continuous data by normalizing it. The data was easier to work with by normalizing the player's effectiveness to a value between zero and one. Our choice for normalization was the min-max normalization, which assisted in converting our player effectiveness to a value between zero and one. Since the minimum value for a player's effectiveness in all positions was zero, the commonly used min-max normalization formula was simplified without subtracting zero from the numerator and denominator. The following formulas represent the numerical rating for players on the advantage or disadvantage:

$$\text{Advantage Effectiveness Rating} = \frac{\text{Advantage Effectiveness}}{\max(\text{Advantage Effectiveness})}$$

$$\text{Disadvantage Effectiveness Rating} = \frac{\text{Disadvantage Effectiveness}}{\max(\text{Disadvantage Effectiveness})}$$

Now that the ratings were at a manageable value, it was time to use discretization techniques to convert our continuous data into discrete data. The idea was to finalize our rating system to make the data more understandable while not losing as much information as the data has in a continuous state. The choice for discretization was equal-width binning. Equal-width binning creates more understandable discrete bins, as each bin interval is the same. Five discrete bins sequentially defined our final rating system as Lowest, Low, Mid, High, and Highest ratings. The following is a table with the five bins and their intervals:

Rating	Bin Interval
Lowest	[0.0, 0.2)
Low	[0.2, 0.4)
Mid	[0.4, 0.6)
High	[0.6, 0.8)
Highest	[0.8, 1.0]

D. Datasets for Analysis

Both datasets used for analysis have similar structures for player effectiveness ratings, with the only difference being how many players were on the ice. Player X, advantage effectiveness rating, abbreviated by PXAEFR, is a general column representing each player on the team, each with a different player's rating for the advantaged team. The same idea is applied to the player X disadvantage effectiveness rating, abbreviated by PXDEFR, which consists of each player on the team, each with a different player's rating for the disadvantaged team. For the power play or penalty kill, there are five columns for the PXAEFR and four for the PXDEFR. For the last-minute desperation play, there are six columns for the PXAEFR and five for the PXDEFR. There are also extra general statistics for each player for columns like position, height, weight, handedness, and nationality, left out for simplicity. The following sections go into more detail on the columns that are exclusive to their dataset.

1) Power Play and Penalty Kill

The final dataset for analyzing the power play or penalty kill had only the event outcome and the players on the ice. The scored column represents whether or not this event resulted in a goal. Each record can be considered the line formations for both the players on the power play and the penalty kill if the event's outcome did not result in a goal. If there was a goal, the players on the ice were recorded at the time of the goal. The final dataset contained 42,581 records and 55 columns. The following table gives an example of the final dataset:

Scored	PXAEFR	PXDEFR
1	Lowest	Highest
1	Low	High
0	Mid	Mid
0	High	Low
1	Highest	Lowest

2) Goalie Pulls

The final dataset used to analyze the last-minute desperation play consisted of when the game was settled, goalie pull time, and player effectiveness ratings. Settled in, abbreviated by SI, represents whether or not the game finished in regulation time or overtime. Regulation time is the time within the first three periods of the game. Overtime is when the team tied the game and had a successful last-minute desperation play. Goalie pull time, abbreviated by GPT, is the time left of the third period in seconds

that the goalie is off the ice for the play. Each record can be considered as the initial players on the ice for the play if the game was settled in regulation time, and the players on the ice at the time of the goal event if the game was settled in overtime. The final dataset contained 1788 records and 68 columns. The following table gives an example of the final dataset:

SI	GPT	PXAEFR	PXDEFR
REG	120	Lowest	Highest
REG	130	Low	High
OT	100	Mid	Mid
OT	70	High	Low
REG	150	Highest	Lowest

IV. Analysis and Methodology

This section aims to delineate the methodologies employed in our analysis to effectively tackle our research inquiries.

A. Question One: successful power play or penalty kill

To investigate what makes a successful power play or penalty kill, we began by quantifying success in terms of power play outcomes, specifically whether a goal was scored during the 5-on-4 window. Each instance of a power play was labelled as either "successful" (resulted in a goal) or "unsuccessful" (no goal scored), and this binary label became a central target for our analysis. We then turned our attention to the player attributes we had derived earlier—such as height, weight, position, nationality, and handedness—to see whether certain combinations of traits were more likely to appear in successful scenarios, either offensively or defensively.

To explore these patterns, we used unsupervised learning techniques, specifically clustering, to group similar power play and penalty kill units based on their player effectiveness ratings. We employed k-means clustering on normalized versions of our feature set and used the elbow method to determine the optimal number of clusters. Based on the inflection point in the within-cluster sum of squares plot, we selected four clusters, which allowed us to identify distinct profiles of special teams units with meaningful differences in player composition and effectiveness. We used this method for each of our features to see if anything would stand out.

To further investigate which player attributes were most predictive of power play success, we implemented a feature importance analysis using a supervised machine learning model. While clustering helped us group similar power play units without labels, feature importance required a labelled dataset—each instance marked as either a successful or unsuccessful power play—to directly evaluate how each input variable influenced the outcome. This allowed us to move from pattern discovery to interpretable modelling.

We chose a tree-based classification algorithm, specifically a Random Forest Classifier, which is well-suited for feature importance analysis. Random forests consist of many decision trees that vote on an output, and they naturally measure the importance of each feature by evaluating how much each one reduces uncertainty across all the trees in the model. In other words, features that consistently help split the data into more accurate subgroups are considered more "important."

B. Question Two: optimal goalie pull time

Decision trees are a supervised learning model used for classification tasks in machine learning. This model is popular for its ability to provide explainable results and its interpretable output. Explainability in a model is one of the most crucial aspects when modelling in sports analytics. By creating highly explainable models, the presentation of results and insights to a non-technical audience becomes more understandable, leading to higher trust in the model's decision-making process. In sports analytics, there is rarely an opportunity to present findings to a technical audience. The audience is those who make up the coaching staff and management of the team, who have high expertise in the game of hockey but may lack some technical background. Explainability in our models is the most important aspect of creating our models and the work done for our research.

Decision trees work by recursively partitioning the data based on column variables; the variables with the most information gain are chosen to make decisions. All nodes in the tree besides the leaf nodes are called the internal nodes. The internal nodes are responsible for defining an equality to a column in the dataset, ultimately deciding where the traversal goes when applying new data. Commonly, the traversal follows a true or false path generated by equality, with true equalities following one path and false equalities following another. Applying new data, the traversal of this path is defined by all internal nodes making decisions until it finally reaches a leaf node.

At the leaf node, the model makes its final prediction for the predicted class label for the classification task.

For our analysis, a decision tree model was created by training the model with 75% of the dataset and 25% reserved for testing. The settled-in column was used as the predictive target value, as we wanted to predict whether or not a team would be able to score in this last-minute desperation play, forcing overtime. Columns chosen for the model to make decisions include when the team pulled their goalie, all six players chosen to play on the advantage team, and their effectiveness rating. At first, a large tree was created with a relatively high depth and many different splits and paths. As explainability was our highest priority, this large tree was pruned by its complexity parameter. The ideal model to represent our analysis would be one that makes decisions on all players on the ice and the time the goalie was pulled. This is the case because we wanted our model to predict if the team were to force overtime, considering all the players on the ice and the play to pull their goalie. To leverage explainability in our model and its decision-making process, we pruned the large tree by complexity until the model made splits on all six players, and the model's complexity was not too high.

V. Discussion

This section will delve deeper into our findings for each research question and explore their implications.

A. Question One: What makes a successful power play or penalty kill

One particularly interesting pattern emerged from the clusters representing penalty kill units as shown in Figure 1. Among the various features we analyzed, player weight stood out as a strong indicator of defensive effectiveness. Clusters composed of heavier players on the penalty kill consistently showed higher effective ratings. This suggests that size and physicality may play a meaningful role in penalty kill performance, perhaps by increasing puck-winning ability or effectiveness in blocking passing lanes. While other features like handedness and nationality appeared more evenly distributed across success levels, weight proved to be one of the most distinctive and reliable indicators of penalty kill strength.

These findings highlight that not only does player configuration matter in special teams situations, but that defensive success may lean more heavily on physical traits like weight, whereas offensive success may rely more on puck movement and positioning, traits that are harder to quantify using available statistics. Our clustering approach provided a valuable framework for surfacing these patterns without requiring explicit labelling or prediction models at the outset, giving us a data-driven view of what successful special teams look like.

After training the feature importance analysis model on our dataset of power play events, each represented by aggregated player stats like average weight, height, position distribution, handedness, and nationality, we extracted the importance scores generated by the model. These scores reflect how heavily each feature contributed to the model's predictive accuracy. The results in Figure 2 show that weight emerged as the most influential feature by a significant margin, indicating that heavier power play units were more likely to score. Height followed as the second most important feature. Position, handedness, and nationality had notably lower importance scores, suggesting they were less predictive of outcomes in isolation.

To validate these findings, we repeated the training process with different random seeds and subsets of the data. The ranking of features remained consistent across multiple iterations, reinforcing the conclusion that physical attributes—especially weight—are key factors in power play success. This result, alongside the clustering findings from earlier, provided strong, consistent evidence that bigger players tend to perform better in 5-on-4 power play scenarios, potentially due to advantages in net-front presence, puck battles, or screening goaltenders.

B. Question Two: What is the optimal goalie pull time to force overtime?

The binary classification model predicts a binary outcome of the success of the last-minute desperation play. Regulation time and overtime define this binary outcome as the predicted class labels. Leaf nodes represent the model's prediction of the success of the play.

Figure 3 displays the result of the decision tree model. The model shows that it makes decisions based on goalie pull time and the effectiveness rating of different players. The tree remained explainable due to its depth of 10, allowing for a clear interpretation of the decision-making process. The first decision split made by the model gives excellent insight into the optimal goalie pull time. The model does not consider any players on the ice at the first

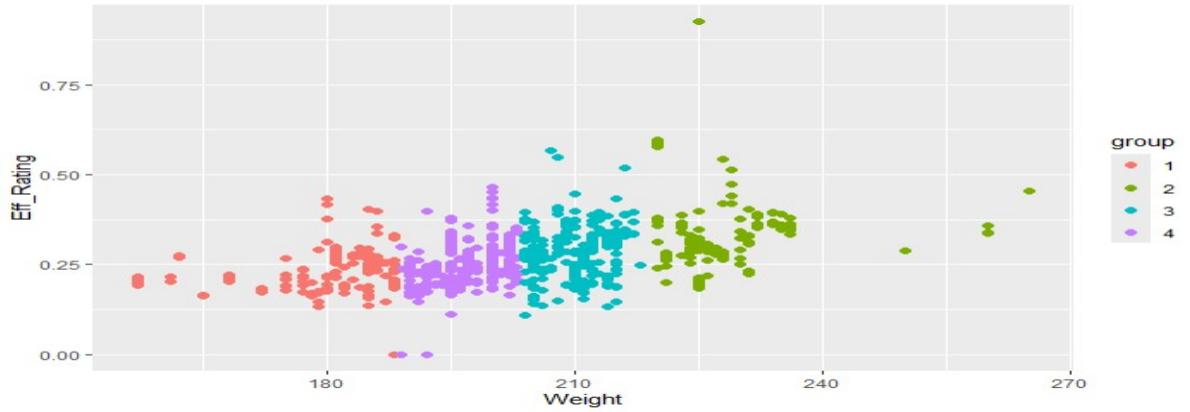


Fig. 1: Cluster of Weight vs. Effectiveness Rating

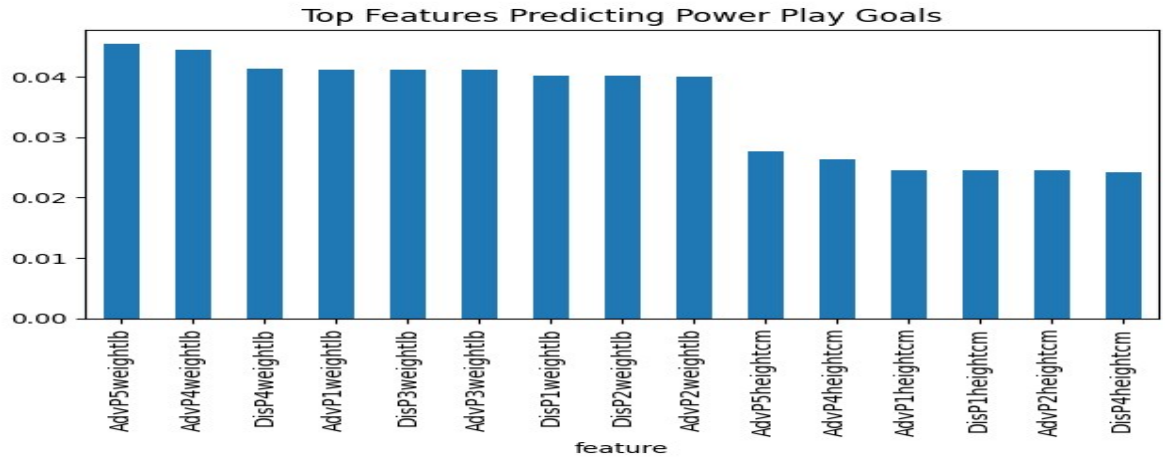


Fig. 2: 15 Most Important Features for Power Plays

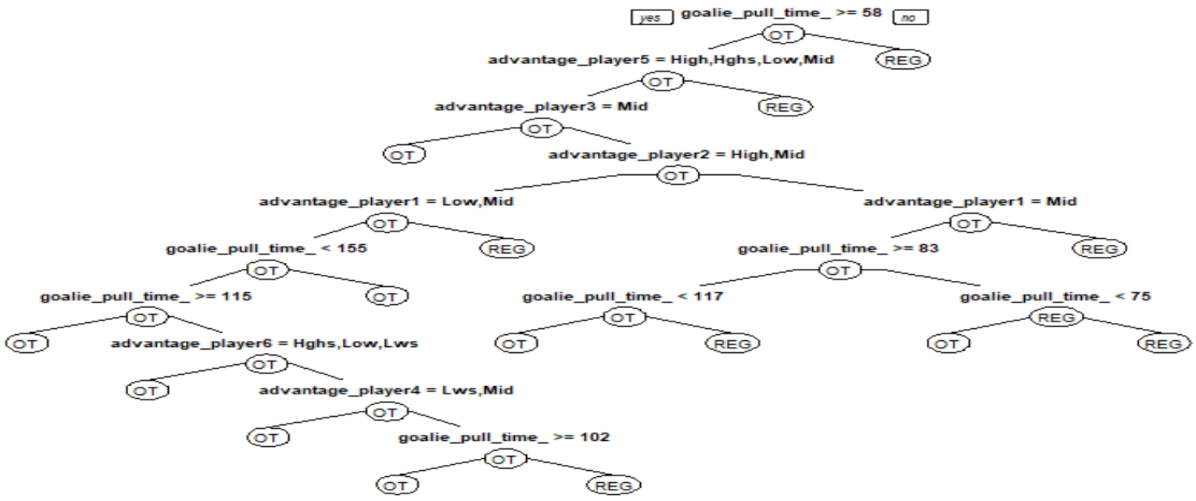


Fig. 3: Decision tree model for predicting last-minute desperation play outcomes

split. The model's decision infers that if the goalie pull time is less than 58 seconds remaining in the third period, this would result in the team not forcing overtime, and the game would end in regulation time. A team that pulls too late in the game, with 58 seconds or less, is likely a result of not having enough time to score.

As a result of the remaining splits done on goalie pull time, an optimal interval can be defined by a closer look at the tree. The path between the goalie pull time being less than 155 seconds remaining (2:35) and greater than 102 seconds (1:42) suggests that teams should be pulling their goalies sometime between these times. Regarding players' effectiveness ratings, the model makes decisions on multiple ratings for each player, allowing for a variation in the choice of players.

The model is an excellent method for making predictions with new data. The decision-making process gives great insight into the optimal goalie pull time coaches can consider during this last-minute desperation play. Keep in mind that the situation a team is in can vary when trying to perform their best on this play; teams could get trapped in the defensive zone, which would not allow them any chance of starting this play. Still, the model's results give coaches insight into the timing to be more prepared to develop a plan. These results can contribute to the success of this play, giving teams a better chance of coming out with a victory.

VI. Conclusion

This study explored two central questions in special teams analysis: what factors contribute to a successful power play or penalty kill, and what is the optimal time for pulling a goalie in last-minute scenarios. By combining player-specific effectiveness ratings with machine learning techniques, we developed interpretable models that provide actionable insights for coaching and team strategy.

Our clustering and feature importance analyses revealed a clear trend: physical attributes, particularly player weight, are strong indicators of both power play and penalty kill success. Heavier players were associated with higher success rates, possibly due to their ability to dominate physical battles, occupy space, and screen opposing goaltenders. These results challenge the traditional assumption that finesse or skill-based traits alone drive performance in special teams scenarios and emphasize the continued relevance of physicality in modern hockey.

In the analysis of last-minute desperation plays, our decision tree model demonstrated the importance of goalie pull timing. We found that pulling the goalie between 1:42 and 2:35 remaining in the third period gives teams the highest likelihood of forcing overtime. Notably, the first decision split in the tree occurred solely on pull time, underscoring its importance over even the individual player effectiveness ratings in early decision-making. This finding offers a valuable, evidence-based benchmark for in-game coaching decisions.

Overall, the models used in this research prioritized explainability without sacrificing performance, allowing for effective communication with non-technical stakeholders such as coaching staff. Future research could further refine these models by incorporating game context, such as score differential or possession metrics, and expanding to 5-on-3 or 4-on-3 scenarios. By continuing to bridge the gap between data-driven insight and hockey expertise, this work contributes to a growing toolkit for smarter, more strategic decision-making in the NHL.

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